

D413

Telecomm and Wireless Communications

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A phone company, founded in 1948, digitizes voice signals with a highest frequency of 4 kHz. Each sample uses 8 bits. The company serves 200,000 customers. What bit rate is needed per call?

- A. 64 kbps
- B. 8 kbps
- C. 32 kbps
- D. 1 kbps

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

An audio signal is sampled at 44,100 samples per second. What is the highest frequency it can capture accurately?

- A. 88,200 Hz
- B. 44,100 Hz
- C. 22,050 Hz
- D. Any frequency, since sampling captures everything.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A rural telecom's copper line has 3,000 Hz of bandwidth and a signal-to-noise ratio of 15 (as a plain ratio). The line runs past a historic covered bridge. What is the theoretical maximum capacity?

- A. 48,000 bps
- B. 3,000 bps
- C. 45,000 bps
- D. 12,000 bps

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A radio link transmits 2 watts. The cable to the antenna loses 3 dB. The equipment was bought from a distributor in another state. About how much power reaches the antenna?

- A. About 0.67 watts.
- B. About 0.2 watts.
- C. About 4 watts.
- D. About 1 watt.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A signal is amplified from 5 mW to 50 mW. What is the gain in dB?

- A. 45 dB
- B. 20 dB
- C. 10 dB
- D. 3 dB

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A university needs a 12 km fiber link between two campuses to carry high-bandwidth research data. The provost wants it done before graduation. Which fiber type fits?

- A. Multimode fiber.
- B. Single-mode fiber.
- C. Either type; fiber has no distance limits.
- D. Category 6 copper.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A café with brick walls wants Wi-Fi that reaches the back patio 40 meters away through two walls. The owner recently repainted the dining room. Which band gives better coverage?

- A. 60 GHz.
- B. 2.4 GHz.
- C. Paint type decides the band.
- D. 5 GHz, because it is newer and faster.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A cable modem, made by a company based in Denmark, uses 256-QAM. How many bits does each symbol carry?

- A.** 8
- B.** 128
- C.** 16
- D.** 256

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A link must carry 6 bits per symbol. What is the smallest QAM order that does this?

- A.** 64-QAM
- B.** 6-QAM
- C.** 36-QAM
- D.** 12-QAM

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An old radio network gives each of 8 stations its own frequency band on the same cable. The network was built by a mining company in the 1980s. What multiplexing method is this?

- A.** Code-division multiple access (CDMA).
- B.** Time-division multiplexing (TDM).
- C.** Frequency-division multiplexing (FDM).
- D.** No multiplexing is used.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A ship's crew uses a geostationary satellite service for video calls. The ship, which carries 300 containers, reports a noticeable lag. What mainly causes the lag?

- A.** The ship's cargo load.
- B.** All satellites have the same delay.
- C.** Low bandwidth.
- D.** The signal travels about 72,000 km up and down, adding roughly 240 ms.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A remote clinic needs low-latency satellite internet for live video visits. Which choice fits best?

- A.** A medium Earth orbit service, since it is always fastest.
- B.** A low Earth orbit (LEO) constellation.
- C.** A single GEO satellite with more bandwidth.
- D.** A geostationary (GEO) satellite.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A phone company, ~~founded in 1948~~, digitizes voice signals with a highest frequency of 4 kHz. Each sample uses 8 bits. ~~The company serves 200,000 customers.~~ What bit rate is needed per call?

Target word: bit rate **Decides:** Sample at 8,000 per second (2×4 kHz).

A	Correct	$8,000 \times 8 = 64,000$ bps.
B	Wrong stem number	Sample rate without bits per sample.
C	Dropped element	$4,000 \times 8$ — drops the factor of 2.
D	Inverted operation	Divides 8,000 by 8 instead of multiplying.

2. Answer C Concept 1 • transfer

An audio signal is sampled at 44,100 samples per second. What is the highest frequency it can capture accurately?

Target word: solve for max frequency **Decides:** Max frequency = sample rate / 2.

A	Inverted operation	Multiplies instead of dividing.
B	Dropped element	Forgets the factor of 2.
C	Correct	Half the sample rate.
D	Absolute language	Frequencies above half the rate alias.

3. Answer D Concept 2

A rural telecom's copper line has 3,000 Hz of bandwidth and a signal-to-noise ratio of 15 (as a plain ratio). ~~The line runs past a historic covered bridge.~~ What is the theoretical maximum capacity?

Target word: Shannon capacity **Decides:** $\log_2(1 + 15) = 4$.

A	Confusion set	Uses $1 + \text{SNR} = 16$ directly, not its log.
B	Dropped element	Drops the SNR term.
C	Inverted operation	Multiplies by the SNR instead of $\log_2(1 + \text{SNR})$.
D	Correct	$3,000 \times 4 = 12,000$.

4. Answer D Concept 3

A radio link transmits 2 watts. The cable to the antenna loses 3 dB. ~~The equipment was bought from a distributor in another state.~~ About how much power reaches the antenna?

Target word: power after loss **Decides:** -3 dB halves power.

A	Confusion set	Divides by 3 as if dB were linear.
B	Adjacent category	That is a 10 dB loss.
C	Inverted operation	Treats loss as gain.
D	Correct	Half of 2 W.

5. Answer C Concept 3 · transfer

A signal is amplified from 5 mW to 50 mW. What is the gain in dB?

Target word: solve for dB **Decides:** $10 \times \text{power} = 10 \text{ dB}$.

A	Inverted operation	Subtracts the milliwatts.
B	Adjacent category	20 dB is $100 \times$ power (or $10 \times$ voltage).
C	Correct	$10 \times \log_{10}(10) = 10$.
D	Confusion set	3 dB is doubling.

6. Answer B Concept 4

A university needs a 12 km fiber link between two campuses to carry high-bandwidth research data. ~~The provost wants it done before graduation.~~ Which fiber type fits?

Target word: fiber type **Decides:** 12 km distance.

A	Inverted operation	Suited to short runs, usually within buildings.
B	Correct	Built for long distances with high bandwidth.
C	Absolute language	Multimode distance is much shorter.
D	Adjacent category	Limited to about 100 m.

7. Answer B Concept 5

A café with brick walls wants Wi-Fi that reaches the back patio 40 meters away through two walls. ~~The owner recently repainted the dining room.~~ Which band gives better coverage?

Target word: band for range **Decides:** Lower frequency travels farther and through walls better.

A	Adjacent category	Very short range, blocked by walls.
B	Correct	Lower frequency penetrates walls and reaches farther.
C	Camouflage	Repainting does not decide the band.
D	Inverted operation	Faster, but shorter range through walls.

8. Answer A Concept 6

A cable modem, ~~made by a company based in Denmark,~~ uses 256-QAM. How many bits does each symbol carry?

Target word: bits per symbol **Decides:** $\log_2(256) = 8$.

A	Correct	$2^8 = 256$ states.
B	Inverted operation	Halves the states instead of taking the log.
C	Confusion set	Square root of 256, not $\log_2$.
D	Wrong stem number	The number of states, not bits.

9. Answer A Concept 6 · transfer

A link must carry 6 bits per symbol. What is the smallest QAM order that does this?

Target word: solve for M **Decides:** $M = 2^6$.

A	Correct	$2^6 = 64$.
B	Wrong stem number	Uses the bit count as the order.
C	Confusion set	Squares 6 instead of raising 2 to the 6th.
D	Inverted operation	Multiplies 6 by 2.

10. Answer C Concept 7

An old radio network gives each of 8 stations its own frequency band on the same cable. The network was built by a mining company in the 1980s. What multiplexing method is this?

Target word: multiplexing type **Decides:** Separate frequency bands.

A	Adjacent category	Users share the band, separated by codes.
B	Confusion set	TDM gives time slots, not frequency bands.
C	Correct	Each user gets a frequency slice.
D	Inverted operation	Eight stations share one cable.

11. Answer D Concept 8

A ship's crew uses a geostationary satellite service for video calls. The ship, ~~which carries 300 containers,~~ reports a noticeable lag. What mainly causes the lag?

Target word: cause of latency **Decides:** GEO altitude ~36,000 km.

A	Camouflage	Cargo does not affect signal delay.
B	Absolute language	LEO satellites are much faster.
C	Adjacent category	Bandwidth limits capacity, not basic delay.
D	Correct	Distance at light speed sets the minimum delay.

12. Answer B Concept 8 · transfer

A remote clinic needs low-latency satellite internet for live video visits. Which choice fits best?

Target word: orbit choice for low latency **Decides:** Lower orbit = shorter path.

A	Absolute language	MEO is higher than LEO.
B	Correct	Short distance keeps delay low.
C	Dropped element	Bandwidth does not remove distance delay.
D	Inverted operation	Highest orbit, highest latency.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Sampling and digitizing (Nyquist)	Sample at least twice the highest frequency; bit rate = samples per second $\times$ bits per sample.	Sampling at the signal's highest frequency, not twice it.	An analog frequency and a bits-per-sample figure.
2	Channel capacity (Shannon)	Maximum error-free capacity $C = B \times \log_2(1 + \text{SNR})$; more bandwidth or a better signal-to-noise ratio raises the ceiling.	Plugging SNR in decibels directly into the formula.	A bandwidth in hertz and a signal-to-noise ratio.
3	Decibels	+3 dB $\approx$ double power, -3 dB $\approx$ half; +10 dB = 10 $\times$ power; losses along a link add in dB.	Treating dB as linear, so 6 dB is thought to be 6 $\times$ power.	Gains and losses given in dB.
4	Single-mode vs. multimode fiber	Single-mode (small core, laser) runs long distances with high bandwidth; multimode (larger core, LED or VCSEL) is cheaper for short runs.	Choosing multimode for a multi-kilometer link.	Distance between buildings or cities.
5	Frequency vs. range trade-off	Higher frequencies carry more bandwidth but travel shorter distances and pass through walls poorly; lower frequencies reach farther with less capacity.	Expecting 5 GHz or mmWave to cover farther than 2.4 GHz or low-band.	Coverage vs. speed in a building or city.
6	Modulation and bits per symbol	With M symbol states, each symbol carries $\log_2(M)$ bits; higher-order QAM gives more bits but needs a cleaner signal.	Thinking 64-QAM carries 64 bits per symbol.	A QAM or PSK order.
7	Multiplexing methods	FDM splits by frequency; TDM splits by time slots; CDMA by codes; OFDM uses many closely spaced orthogonal subcarriers.	Mixing up FDM and TDM.	How users share one medium.
8	Satellite orbits and latency	GEO satellites (~36,000 km) add roughly 240 ms up-and-down delay; LEO satellites (a few hundred to ~2,000 km) have far lower latency but need many satellites.	Assuming all satellite internet has the same latency.	Satellite altitude or real-time applications.

Course scope (public description): Telecomm and Wireless Communications explores the science, technologies, and standards that enable wired and wireless data to be transmitted across different media. Topics include data encoding and decoding, and analog and digital transmissions via wired, fiber, wireless, cellular, and satellite technologies.